

MCA
(SEM IV) THEORY EXAMINATION 2022-23
COMPILER DESIGN

Time: 3 Hours**Total Marks: 70****Note:** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief. 2 x 7 = 14**

- a. Differentiate between Kleen Closure and Positive Closure with suitable examples
- b. Define patterns, lexeme, and tokens?
- c. What do you mean by Cross-Compiler?
- d. Design a DFA for the languages accepting all strings containing maximum three 'a's over the input $\Sigma = \{a,b\}$.
- e. Discuss the process of left factoring removal?
- f. What are shift/ reduce and reduce/reduce conflicts?
- g. Explain dominators and its properties

SECTION B**2. Attempt any three of the following: 7 x 3 = 21**

- a. Define bootstrapping with an example.
- b. Perform Stack Implementation of shift reduce parsing on given string $id1 + id2 + id3$ by using Grammar
 $E \rightarrow E+E \mid E * E \mid E \mid id$
- c. Explain the problems of top- down parsers.
- d. Translate "conversion of expression $a+b*c$ into postfix expression" into Three address code and parse tree using syntax directed translation scheme
- e. Compute FIRST and FOLLOW for given Grammar
 $S \rightarrow ACB \mid CbB \mid Ba$
 $A \rightarrow da \mid BC$
 $B \rightarrow g \mid \epsilon$
 $C \rightarrow h \mid \epsilon$

SECTION C**3. Attempt any one part of the following: 7 x 1 = 7**

- a. Outline the NFA and also give the minimized DFA for the following regular expression $(a/b)^* abb$
- b. Write short notes on symbol table manager and error handler?

4. Attempt any one part of the following:

7 x 1 = 7

- a. Point out the various phases of compiler and Write down the output of each phase of the compilation for the expression $a := (b * c) + (c * 10)$.
- b. What is meant by lexical analysis? Identify the lexemes that makeup the token in the following program segment. indicate the correspond token and pattern.

```
Void swap (int i, int j)
{
  int t; t = i ; i = j ; j = t ;
}
```

5. Attempt any one part of the following:

7 x 1 = 7

- a. What do you mean by operator precedence? Construct the operator precedence table for the following grammar
 $E \rightarrow E+E \mid E-E \mid E * E \mid E / E \mid E^E \mid (E) \mid -E \mid id$
- b. Consider the following grammar and find is it LL(1) or not with explanation for the following grammar:
 $S \rightarrow iCtSeS \mid iCtS \mid a$
 $C \rightarrow b$

6. Attempt any one part of the following:

7 x 1 = 7

- a. Explain syntax directed translation scheme. Also write a syntax directed definition of a simple desk calculator and draw an annotated parse tree for $4 * 3 + 2$
- b. What is three address code and explain different record structure to represent three address code on following example $A := B * (C - D)$

7. Attempt any one part of the following:

7 x 1 = 7

- a. Explain Directed Acyclic Graph and how it is different form Flow graph. Also create a DAG of three address code of following program

```
begin
  MUL=0
  J:=1
do
  begin
    MUL:=MUL+A[J]*B[J]
    J=J+1
  end while J<=10
end
```
- b. Write short note on the following.
 - I. Explain construction of basic block, identification of leader and flow graph.
 - II. Back patching.